Normal distribution probabilities

1 Consider the standard normal distribution tables which give the area between 0 and a given z-score:

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0754
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2258	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2518	0.2549
0.7	0.2580	0.2612	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2996	0.3023	0.3051	0.3079	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4430	0.4441
1.6	0.4452	0.4463	0.4474	0.4485	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4700	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4762	0.4767
2	0.4773	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4980	0.4980	0.9981
2.9	0.4981	0.4982	0.4983	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990

Using the tables, find the area under the normal curve between:

a 1 standard deviation below the mean and 2 standard deviations above the mean

b 1.60 and 1.80 standard deviations above the mean

2 Using the tables from Q2, find the area unde[illegible] normal curve:

a To the left of $z = 1.15$.

b To the right of $z = 1.38$.

c To the left of $z = -1.76$.

d To the right of $z = -1.16$.

e Between $z = -1.11$ and $z = -1.59$.

f Between $z = 1.51$ and $z = 1.89$.

3 Using the tables from Q2, find the percentage, to two decimal places, of the data that is:

a Less than $z = 0.71$.

b Greater than $z = -1.15$.

c Less than $z = 0.18$.

d Between $z = -1.29$ and $z = 2.35$.

e Between $z = 0.39$ and $z = 2.48$.

4 Using the table from Q2, find the probability that a variable has a z-score less than $z = 0.85$. Round your answer to four decimal places.

5 Consider the standard normal curve.

a State the size of the area under the curve that is above the mean.

b State the size of the area under the curve that is below the mean.

6 Find the area under the curve, to four decimal places, for each part of the standardized normal curves described below:

a Between $z = -1.23$ and $z = -1.55$

b To the left of $z = 1.45$

c To the right of $z = 1.58$

d To the left of $z = -1.23$

e To the right of $z = -1.17$

f Between $z = 1.52$ and $z = 1.87$

g Between two standard deviations below the mean and three standard deviations above the mean.

h Between 1.10 and 1.60 standard deviations above the mean.

7 Calculate the percentage of standardized data, to two decimal places, that is:

a Greater than $z = -1.51$

b Between $z = -1.14$ and $z = 2.37$

c Between $z = 0.47$ and $z = 2.46$

8 Calculate the probability, to four decimal places that a z-score is:

a Either at most -1.08 or greater than 2.07

b Greater than -0.63 and at most 1.44

c At most 1.60 given that it is greater than -0.69

d At most 1.03 given that it is less than 2.58

9 Using your calculator, find the value of k to four decimal places for the following probabilities:

a The probability of a z-score being at most k is equal to 0.8031 in the standard normal distribution.

b The probability of a z-score being greater than k is equal to 0.7934 in the standard normal distribution.

c The probability of a z-score being at most k is equal to 0.218 in the standard normal distribution.

d The probability of a z-score being greater than k is equal to 0.1562 in the standard normal distribution.

e The probability of a z-score being greater than $-k$ and at most k is equal to 0.6123 in the standard normal distribution.

f The probability of a z-score being greater than -2.11 and at most k is equal to 0.8273 in the standard normal distribution.

g The probability of a z-score being greater than k and at most 2.45 is equal to 0.7975 in the standard normal distribution.

Applications

10 The mean intelligence quotient (IQ) score is 100, with a standard deviation of 14, and the scores are normally distributed. Given this information, the approximate percentage of the population with an IQ greater than 130 is closest to

A 2% **B** 0% **C** 98% **D** 16%

11 The average life span of a tortoise is estima[illegible] be approximately 100 years, with a standard deviation of 7 years.

a Determine the z-score of a life span of 102 years. Round your answer to two decimal places.

b Find $P\left(X>102\right)$. Round your answer to four decimal places.

c Find the probability of a tortoise living beyond 102 years of age.

12 Research scientists have measured the heights of a large number of maple trees in an area of forest. The average height of these trees is 60 m, with a standard deviation of 6 m.

a Find the z-score of a height of 68 m. Round your answer to two decimal places.

b Find $P\left(X<68\right)$. Round your answer to four decimal places.

c Find the height of a maple tree in the forest that is represented by the probability of 0.9082.

13 A sprinter is training for a national competition. She runs 400 yd in an average time of 75 seconds, with a standard deviation of 6 seconds.

a Determine the z-score of a time of 67 seconds. Round your answer to two decimal places.

b Find $P\left(X<67\right)$. Round your answer to four decimal places.

c Find the time of the sprinter to run 400 yd that is represented by the probability of 0.0918.

14 The lengths of adult male lizards of a particular species are thought to be normally distributed with a mean of 19.5 cm and a standard deviation of 4.5 cm.

a Find the z-score of a lizard length of 19.5 cm.

b Find the probability that a randomly chosen adult male lizard of this species will have a length less than 19.5 cm.

c Find the z-score of a lizard length of 15 cm.

d Find the probability that a randomly chosen adult male lizard of this species will have a length between 15 cm and 19.5 cm. Round your answer to four decimal places.

15 A box of breakfast cereal has "contains 500 [illegible] of breakfast cereal" printed on it. The amount of breakfast cereal contained in these boxes is normally distributed with a mean of 512 grams and a standard deviation of 9 grams.

a Find the z-score of a box containing 500 grams of cereal, to two decimal places.

b Find the probability that a randomly chosen box of this cereal contains less than 500 grams. Round your answer to four decimal places.

c In a random sample of 100 boxes of this cereal, find the approximate number of boxes we should expect to contain less than 500 grams.

16 Pre-bagged packs of bananas are marked as "contains approximately 5 lb". The mass of the contents of such bags is normally distributed with a mean of 5.01 lb and a standard deviation of 0.08 lb.

a Find the z-score of a 5 lb pack of bananas. Round your answer to two decimal places.

b Find the probability that a randomly chosen pack of these bananas has a mass of less than 5 lb. Round your answer to four decimal places.

17 The scaled results in a national mathematics test are normally distributed with a mean of 65 and a standard deviation of 7.

a Find the z-score of a test result of 71. Round your answer to two decimal places.

b Find the probability that a randomly selected candidate who sat this test has a scaled result of more than 71. Round your answer to four decimal places.

c Find the z-score of a test result of 61. Round your answer to two decimal places.

d Find the probability that a randomly selected candidate who sat this test has a scaled result of between 61 and 71. Round your answer to four decimal places.

e The scaled result of 51 is two standard deviations below the mean. Find the probability that a randomly selected candidate who sat this test has a scaled result of less than 51. Round your answer to four decimal places.

18 The mean height of an adult male is 1.78 m, with a standard deviation of 9 cm.

a Find the z-score of a height of 1.69 m.

b If 700 males are chosen at random, find the approximate number males who are taller than 1.69 m.

19 The mean number of cookies in a box is 35, [illegible] a standard deviation of 4.

a Find the z-score of a box containing 27 cookies.

b If 4000 boxes of cookies are produced, find the approximate number of boxes with more than 27 cookies.

20 Assume the mean time a male professional diver can hold his breath is 116 seconds, with a standard deviation of 7 seconds.

a Find the z-score of a time of 102 seconds.

b If 600 male professional divers are selected at random, find the approximate number of divers who can hold their breath for longer than 102 seconds.

21 The grades of an exam recently completed by a class are normally distributed. The mean grade in the exam was 57, with a standard deviation of 4.

a Find the z-score of a grade of 53.

b Find the percentage of students who achieved a grade above 53, to the nearest percent.

c Find the z-score of a grade of 49.

d Find the percentage of students who achieved a grade below 49, to the nearest percent.

22 Blood pressure is one human characteristic that has a normal distribution. That is, high and low values are unlikely, and average values are more likely.
In the general population, the mean diastolic blood pressure is 83 mmHg (millimeters of mercury) and the standard deviation is 16 mmHg.

a Find the z-score for a blood pressure of 110 mmHg, to two decimal places.

b Find the percentage of the population with blood pressure greater than 110 mmHg, to the nearest percent.

c Find the z-score for a blood pressure of 42 mmHg, to two decimal places.

d Find the percentage of the population with blood pressure below 42 mmHg, to the nearest percent.

23 The time from Laura getting out of bed until her arrival at school is normally distributed with a mean of 36 minutes and a standard deviation of 6 minutes. Laura's arrival at school is classified as being late if it occurs after 9:15 am.

a If Laura gets out of bed at 8:29 am, state the number of minutes she has to get to school.

b Find the z-score for the time of 46 minutes, to two decimal places.

c If Laura gets out of bed at 8:29 am, find the probability that she will arrive late. Round

24 A particular vitamin has 42 mg which is equivalent to 124% of the recommended daily intake of that vitamin. A 124 ml box of fruit juice contains approximately 42 mg of this vitamin. Suppose that the mass of the vitamin in the 124 ml box of the juice is normally distributed with a mean 42 mg and standard deviation of 4.5 mg.

a Find 100% of the recommended daily intake of the vitamin, to the nearest milligram.

b Find the z-score of a vitamin mass of 34 mg, to two decimal places.

c Find the probability that a randomly chosen 124 mL box of this fruit juice contains less than the recommended daily intake of the vitamin, to four decimal places.